Introduction to Biophysics II: Quantum physics & biology

Lecture 14: Electron transfer in biology

Reading:
- Quantum Effects in Biology, chapter 9
- Bolton et al.; Electron Transfer in Inorganic, Organic, and Biological Systems. Advances in Chemistry; American Chemical Society: Washington, DC, 1991
- Newton MD, Chem. Rev.91, 767, 1991
- Marcus RA, Sutin N. BBA 811 (1985) 265
- Marcus RA, JCP 24 (1956) 966
- C.C. Page, C.C. Moser, X.X. Chen, P.L. Dutton, Natural engineering principles of electron tunneling in biological oxidation-reduction, Nature 402 (1999) 47–52

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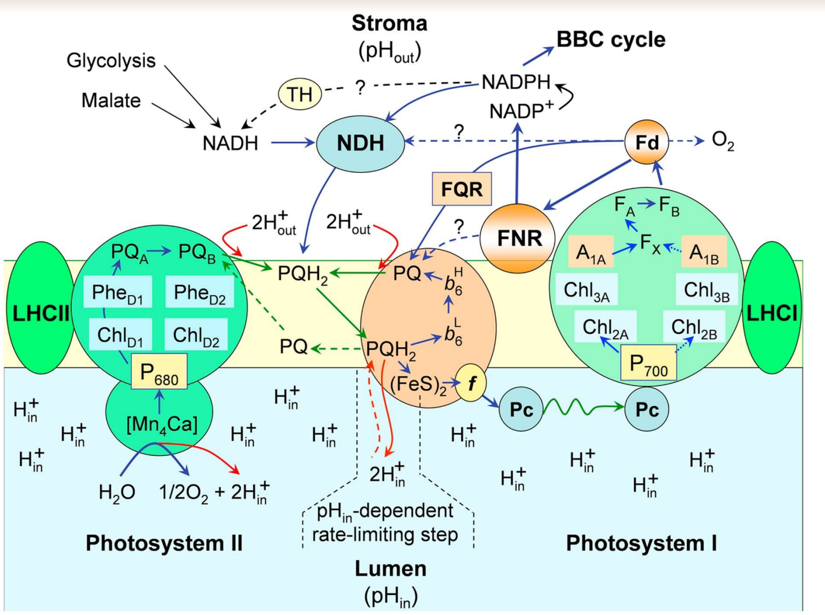
Charge transfer: Photosynthesis



https://www.sciencedirect.com/science/article/pii/S098194281300435X

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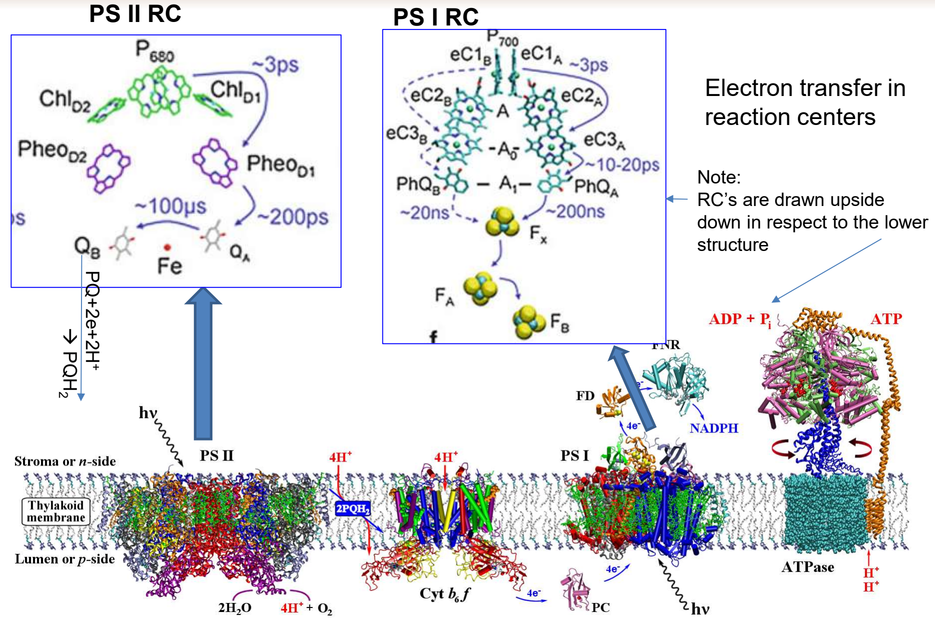
Electron transfer: example



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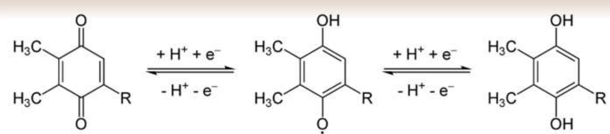
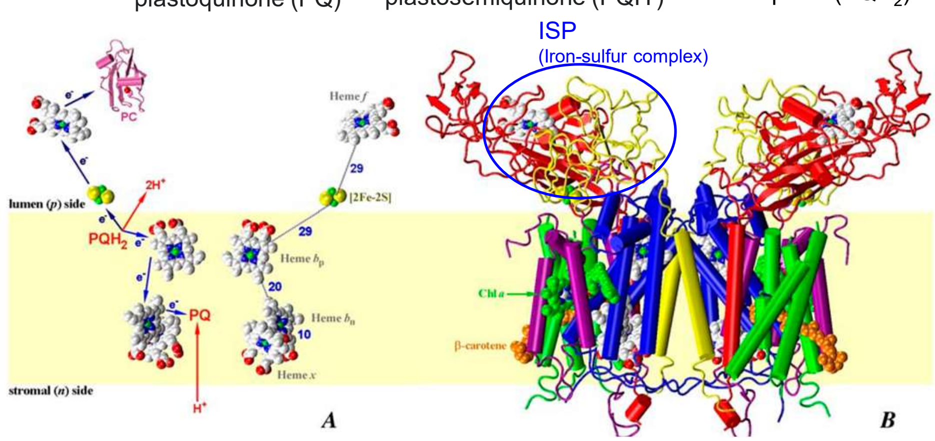
Electron transfer: bifurcated ET



plastoquinone (PQ) plastosemiquinone (PQH·) Plastoquinol (PQH_2)



Cytochrome b_6f

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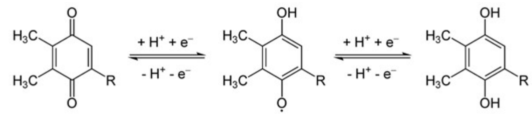
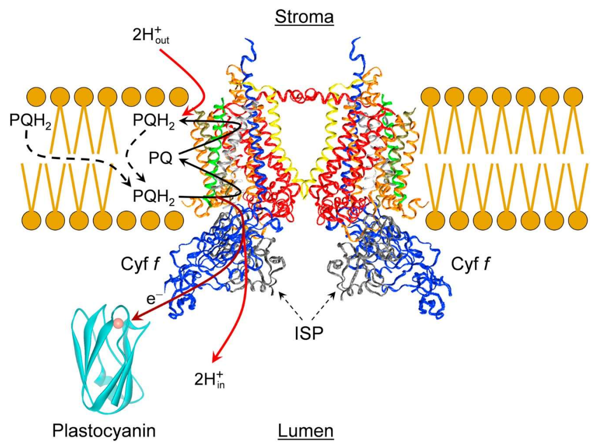
Electron transfer: example

Cyclic bifurcated
electron transfer





Cytochrome b_6f

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Electron transfer: scenarios

Transfer of electron from donor to acceptor:

$D + A \rightarrow D^{+} + A^{-}$

Transfer of electron from excited donor to neutral acceptor:

$D^{*} + A \rightarrow D^{+} + A^{-}$

Transfer of electron from reduced donor to neutral acceptor:

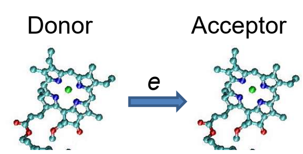
$D^{-} + A \rightarrow D + A^{-}$

Transfer through a chain of cofactors:

$D^{*} + A1 + A2 + \ldots \rightarrow D^{+} + A1^{-} + A2 + \ldots \rightarrow D^{+} + A1 + A2^{-} + \ldots$

Recombination:

$D^{+}A^{-} \rightarrow D\,A$



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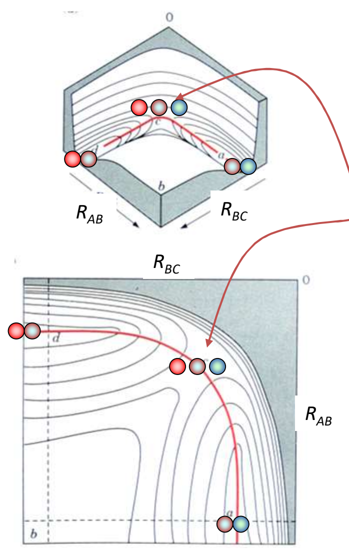
Recall: biophysics I

Transition state: classical thermodynamics

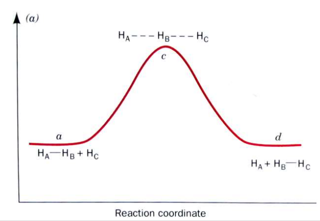
Suppose it is reaction: $H_A\text{-}H_B + H_C \rightarrow H_A{\cdots}H_B{\cdots}H_C \rightarrow H_A + H_B\text{-}H_C$



The potential energy of $H_A{\cdots}H_B{\cdots}H_C$ as a function of distances between hydrogens A-B and B-C

The highest point on the lowest path from one state to another: *transition state*. Complex in this state is called *activated*.

Concentration of *active complex* is small, and the decomposition of activated complex is the rate-determining process of the reaction



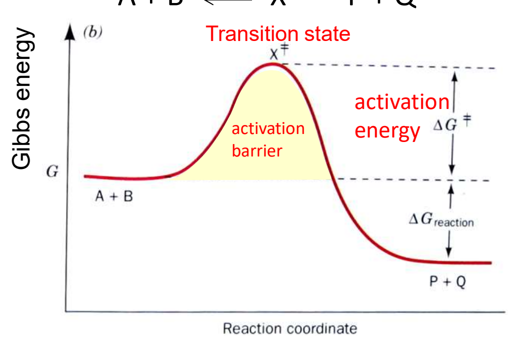
Recall: biophysics I

Electron transfer: classical

Chemical reaction rate (classical):

$$A + B \xrightarrow{k} P + Q$$

$$A + B \rightleftharpoons X^{\ddagger} \xrightarrow{k'} P + Q \quad \text{heat}$$



$$\frac{d[\mathrm{P}]}{dt} = k[\mathrm{A}][\mathrm{B}] = k'[\mathrm{X}^{\ddagger}]$$

k - ordinary rate constant

k' - decompos. of $X^{\ddagger}$

$$k = k'\frac{[\mathrm{X}^{\ddagger}]}{[\mathrm{A}][\mathrm{B}]}$$

$$K^{\ddagger} = \frac{[\mathrm{X}^{\ddagger}]}{[\mathrm{A}][\mathrm{B}]} = e^{-\Delta G^{\ddagger}/kT}$$

$$\boxed{k = k' e^{-\Delta G^{\ddagger}/kT}}$$

Arrhenius equation (1889)

Decomposition is assumed to happen in time of one thermal vibration ν with probability κ to produce product:

$$\boxed{k = \kappa \nu e^{-\Delta G^{\ddagger}/kT}}$$

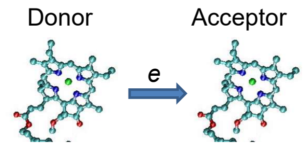
Can also us $E=h\nu$ and $E=kT$ → $\nu=kT/h$ → $k = \frac{kT}{h} e^{-\Delta G^{\ddagger}/kT}$ (assuming κ=1)

R. Marcus (1923-)

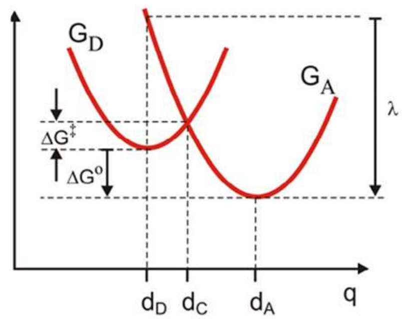
Electron transfer: Marcus theory

Marcus theory, developed in 1956-1965, *Nobel prize 1992*
classical approach
(non-adiabatic)
Transfer of electron from donor to acceptor :

$D + A \rightarrow D^{+} + A^{-}$



First, we assume that the potential energy is represented by parabolas (harmonic oscillator):
G_D – Potential (Gibbs) energy of state when an electron is on donor
G_A –Potential energy when an electron is on the acceptor
ΔG_0 – difference in energies of initial and final states (relaxed)
q – generalized reaction coordinate
$\Delta G^{\ddagger}$ - activation barrier
λ - reorganizational energy



Marcus theory: pages are adopted from:
Download Center https://chem.libretexts.org/@go/page/134526 (accessed Mar 14, 2021).

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Electron transfer…

Marcus: classical approach (non-adiabatic)
Parabolas (harmonic oscillator):

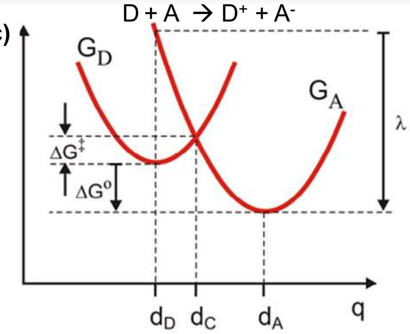
Following Arrhenius equation for transition state theory:

$$\boxed{k = \kappa \nu e^{-\Delta G^{\ddagger}/kT}}$$

The probability κ is here the probability for an electron to transfer/tunnel from donor to acceptor, denoted as k_{el}:

$$k = k_{el} \nu e^{-\Delta G^{\ddagger}/kT}$$

$D + A \rightarrow D^{+} + A^{-}$



Note: $\Delta G^0 < 0$

ν Is the frequency of passage (nuclear motion) of the transition state at $q = d_C$.
In classical treatment electron transfer k_{el} is taken to be unity (as κ)
– not a valid approach for electron transfer!

Need to know $\Delta G^{\ddagger}$

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Electron transfer…

Marcus: classical approach (non-adiabatic)
Parabolas (harmonic oscillator):

$$G_D(q) = \frac{1}{2}m\omega_0^2(q - d_D)^2$$

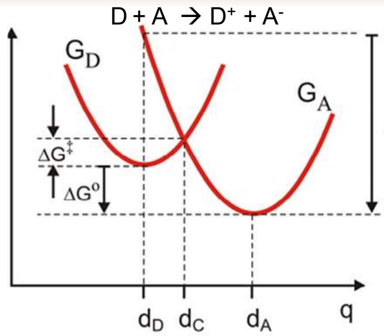
$$G_A(q) = \frac{1}{2}m\omega_0^2(q - d_A)^2 + \Delta G^0$$

The barrier height at the crossing point d_C:

$$G_D(d_C) = G_A(d_C)$$

$$\frac{1}{2}m\omega_0^2(d_c - d_D)^2 = \Delta G^\circ + \frac{1}{2}m\omega_0^2(d_C - d_A)^2$$

$$d_C = \frac{\Delta G^\circ}{m\omega_0^2}\left(\frac{1}{d_A - d_D}\right) + \frac{d_A + d_D}{2}$$



Note: $\Delta G^0 < 0$

Note that → $\lambda = G_A(d_D) - G_A(d_A) = \frac{1}{2}m\omega_0^2(d_D - d_A)^2$
And we get:

$$d_C = \frac{\Delta G^\circ}{2\lambda}(d_A - d_D) + \frac{d_A + d_D}{2}$$

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Electron transfer…

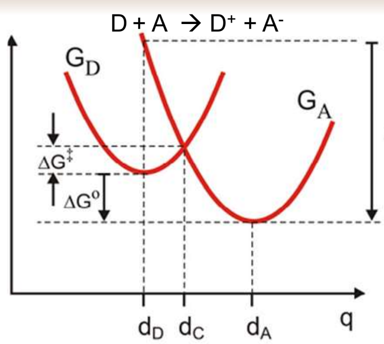
Marcus: classical approach (non-adiabatic)

Have: $d_C = \frac{\Delta G^\circ}{2\lambda}(d_A - d_D) + \frac{d_A + d_D}{2}$

$$\lambda = \frac{1}{2}m\omega_0^2(d_D - d_A)^2$$

And the barrier height:

$$\begin{aligned}\Delta G^\dagger &= G_D(d_C) - G_D(d_D)\\ &= \frac{1}{2}m\omega_0^2(d_C - d_D)^2\\ &= \frac{1}{4\lambda}[\Delta G^\circ + \lambda]^2\end{aligned}$$



$$G_D(q) = \frac{1}{2}m\omega_0^2(q - d_D)^2$$

Plug this into Arrhenius equation: $k = k_{el}\nu e^{-\Delta G^\ddagger / kT}$

And we get $\boxed{k_{ET} = k_{el}\nu \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)}$

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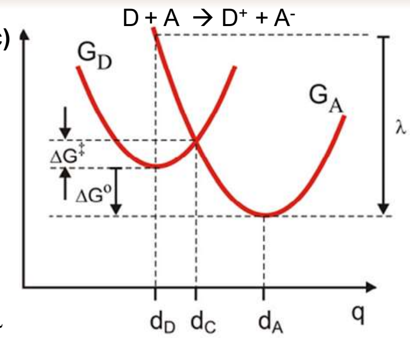
Electron transfer…

Marcus: classical approach (non-adiabatic)

$$\boxed{k_{ET} = k_{el} \nu \exp\left(-\frac{\left[\Delta G_0 + \lambda\right]^2}{4\lambda kT} \right)}$$



- Electron transfer rate depends exponentially on temperature
- Electron transfer maximizes when $\Delta G^0 = -\lambda$

k_{el} = ? → electron transfer involves *tunneling* from one location to another.

Note: Distance between donor and acceptor, R_{DA}, is not equal to d_A-d_D!!!

q is the “reaction coordinate” that involves a complex trajectory in a multidimensional space comprised of coordinates of *all atoms* in the system that shift as the result of electron transfer
R is the *actual* distance between the donor and acceptor to be tunneled by the electron

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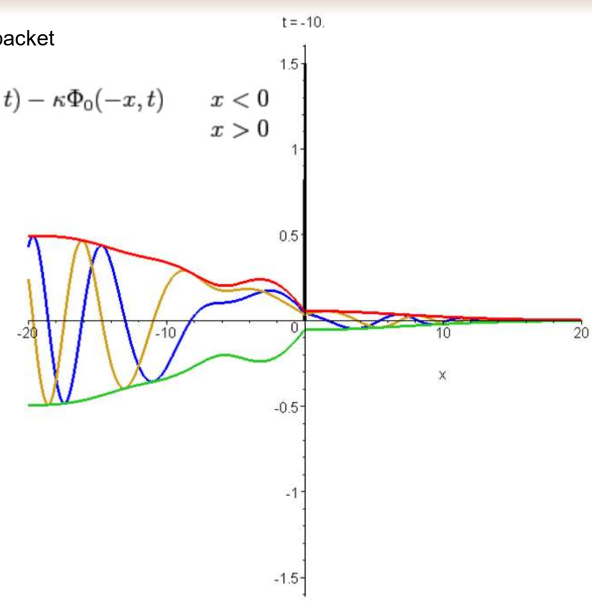
Tunneling

Free space Gaussian wavepacket

$$\Phi(x,t) = \begin{cases} (-\partial_x + \kappa)\Phi_0(x,t) - \kappa\Phi_0(-x,t) & x < 0 \\ -\partial_x \Phi_0(x,t) & x > 0 \end{cases}$$



https://www.victoria.ac.nz/scps/about/staff/johnlekner

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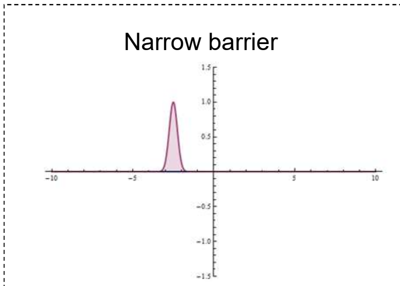
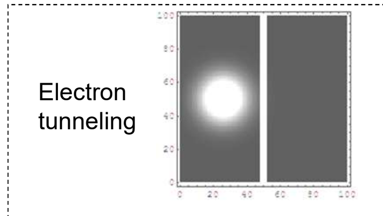
Tunneling





http://en.wikipedia.org/wiki/Quantum_tunnelling

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Tunneling: square barrier

Tunneling through a square barrier.

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

$$V(x) = \begin{cases} 0, & \text{within free space} \\ V_0, & \text{within barrier} \end{cases}$$

Solve SE: $\hat{H}\psi = E\psi \Rightarrow \frac{\partial^2 \psi}{\partial x^2} = -2m\hbar^{-2}(E - V)$

$$\frac{\partial^2 \psi_{1,2}}{\partial x^2} = -2mE\hbar^{-2}\psi_{1,2}, \quad \text{within free space}$$

$$\frac{\partial^2 \psi_3}{\partial x^2} = -2mE\hbar^{-2}(E - V_0)\psi_3, \quad \text{within barrier}$$

Harmonic wave

$$\psi_1 = Ae^{ikx} + Be^{-ikx}$$

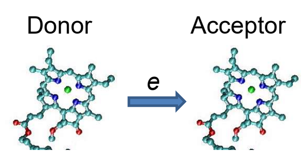
$$\psi_2 = Ce^{ikx} + De^{-ikx}$$

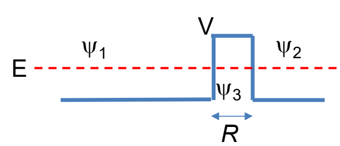
$$k = \sqrt{2mE} / \hbar$$

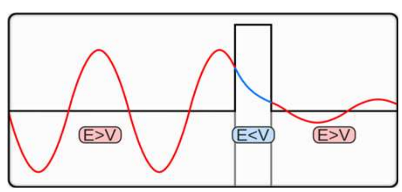
$$\kappa = \sqrt{2m(V_0 - E)} / \hbar$$

$$\psi_3 = Fe^{\kappa x} + Ge^{-\kappa x}$$

Exponential decay







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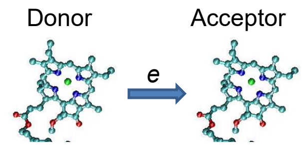
Tunneling: square barrier

Tunneling through square barrier.

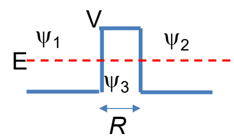


$$\psi_1 = Ae^{ikx} + Be^{-ikx}$$
$$\psi_2 = Ce^{ikx} + De^{-ikx}$$
$$\psi_3 = Fe^{\kappa x} + Ge^{-\kappa x}$$

$$k = \sqrt{2mE}/\hbar$$
$$\kappa = \sqrt{2m(V_0 - E)}/\hbar$$

1. Boundary conditions: functions and their first derivatives of x must be continuous.
2. We are interested in coefficients A (moving right) and C (moving right)
3. Transmission:

$$\psi_1(0) = \psi_3(0)$$
$$\partial\psi_1(0)/\partial x = \partial\psi_3(0)/\partial x$$
$$\psi_3(R) = \psi_1(R)$$
$$\partial\psi_3(R)/\partial x = \partial\psi_1(R)/\partial x$$



$$T = \left|\frac{C}{A}\right|^2 = \left|\frac{4i\kappa m e^{-ikR-mR}}{(k+i\kappa)^2 - (k-i\kappa)^2 e^{-2\kappa R}}\right|^2 = \frac{1}{1 + V_0^2 \sinh^2(\kappa R)/\left[4E(V_0 - E)\right]}$$

For high and/or wide barrier (κR>>1): $\sinh x = \frac{e^x - \cancel{e^{-x}}}{2}$

$$T = \frac{16E(V_0 - E)}{V_0^2}\exp(-2\kappa R) = \frac{16E(V_0 - E)}{V_0^2}\exp\left(-\frac{2R}{\hbar}\sqrt{2m(V_0 - E)}\right)$$

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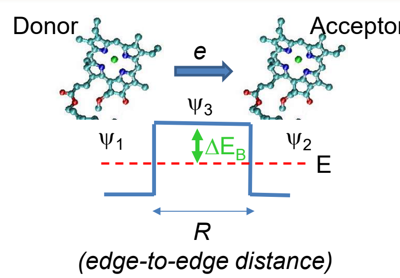
Tunneling: square barrier

Tunneling through high/wide square barrier:

$$T = \frac{16E(V_0 - E)}{V_0^2}\boxed{\exp\left(-\frac{2R}{\hbar}\sqrt{2m(V_0 - E)}\right)}$$



Denote $\Delta E_B = V_0 - E$, rewrite:

$$T = \frac{16E\Delta E_B}{V_0^2}\exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta E_B}\right)$$

Tunneling depends exponentially on the distance and square root of barrier height. Exponential term dominates transmission rate at large distances:

$$T \propto \exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta\bar{E}_B}\right)$$ (where $\Delta\bar{E}_b$ is average barrier height for uneven barriers)

And we can rewrite $k_{ET} = k_{el}\nu\exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$

$$\boxed{k_{ET} \propto \exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta\bar{E}_b}\right)\exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)}$$

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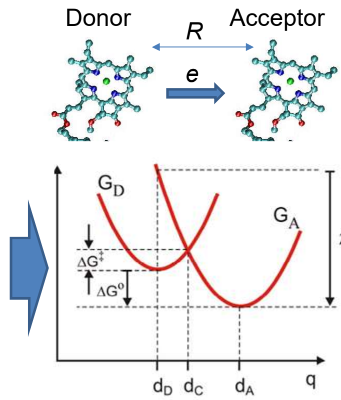
Activation barrier + tunneling

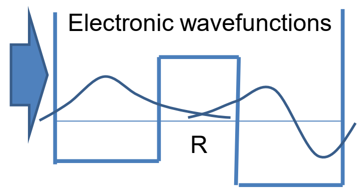
Electron transfer:

$$\boxed{k_{ET} \propto \exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta\bar{E}_b}\right)\exp\left(-\frac{[\Delta G_0+\lambda]^2}{4\lambda kT}\right)}$$



$\boxed{\exp\left(-\frac{[\Delta G_0+\lambda]^2}{4\lambda kT}\right)}$ The second term is a requirement that the energies of initial and final states match!
This is probability that electronic+vibrational energy of donor and acceptor cross.
ΔG‡ - is activation barrier for nuclear reaction.
q is the reaction coordinate

$\boxed{\exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta\bar{E}_b}\right)}$ The first term is due to electron tunneling, ΔE_b is barrier height for electron tunneling!
R is actual distance along the line connecting donor and acceptor.



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Electron transfer…

Sometimes it is expressed in terms similar to quantum mechanical approach:

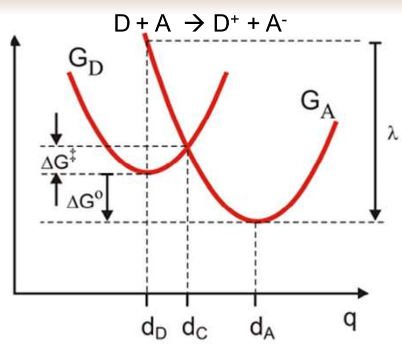
$$k_{ET} = \frac{2\pi}{\hbar} H_{DA}^2 \rho_{FC}^{clas}$$

H_{DA}^2 Tunneling matrix element,

$$H_{DA} \propto \sqrt{T} \propto \exp\left[-\left(\frac{\sqrt{2m_e\Delta\bar{E}_b}}{\hbar}\right)R_{DA}\right]$$

ρ_{FC}^{clas} Classical Boltzmann activation factor ("classical Franck-Condon factor")

$$\rho_{FC}^{clas} = \frac{1}{\sqrt{2\pi\sigma_{\Delta G}^2}}\exp\left(\frac{-[\Delta G_0+\lambda]^2}{4\lambda kT}\right)$$



$\sigma_{\Delta G}^2$ - Variance in the energy gap G_D-G_A:

$$\sigma_{\Delta G}^2 = 2\lambda kT$$

$$\boxed{\ln k_{ET} = const - \left(\frac{2\sqrt{2m_e\Delta G^{\ddagger}_{average}}}{\hbar}\right)R_{DA} - \frac{[\Delta G_0+\lambda]^2}{4\lambda kT}}$$

From: Quantum effects in biology

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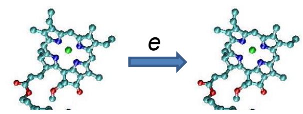
Electron transfer

Quantum-mechanical approach

Transfer of electron from reduced donor to neutral acceptor:

$D^- + A \rightarrow D + A^-$

Donor Acceptor



D$^-$ (*e* on donor) $\psi_D = \phi_D \chi_D \sigma_D$ **A**$^-$ (*e* on acceptor) $\psi_A = \phi_A \chi_A \sigma_A$

Consider wave function of electron on donor (initial state) and on acceptor (final): (Born-Oppenheimer approximation)

Construct superposition state for the electron:

$$\Psi = C_D \phi_D \chi_D \sigma_D + C_A \phi_A \chi_A \sigma_A$$

ϕ - spatial wave function
χ - nuclear wave function
σ - spin

Initial state: C_D*=1*, C_A*=0*.

Need to solve time-dependent Schrödinger equation to find $C_A(t)$: $\hat{H}\Psi = i\hbar \frac{\partial \Psi}{\partial t}$

The solution is mathematically similar to Förster/Dexter energy transfer:

Recall Lecture 6

$$C_2(\tau) = H_{21}(1-\exp[i(E_2-E_1)t/\hbar])/(E_2-E_1)$$

$\boxed{C_A}$ $\langle \psi_2 | \tilde{H}_{21} | \psi_1 \rangle \leftarrow \boxed{\langle \phi_A \chi_A \sigma_A | H_{AD} | \phi_D \chi_D \sigma_D \rangle}$

$\boxed{E_1 = E_D \quad E_2 = E_A}$

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Electron transfer

$D^- + A \rightarrow D + A^-$

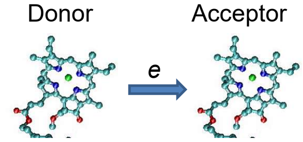
Quantum-mechanical approach

$$\Psi = C_D \phi_D \chi_D \sigma_D + C_A \phi_A \chi_A \sigma_A$$

Initial state: C_D*=1*, C_A*=0*:

$$\langle \phi_A \chi_A \sigma_A | H_{AD} | \phi_D \chi_D \sigma_D \rangle$$

Donor Acceptor



$\psi_D = \phi_D \chi_D \sigma_D$ $\psi_A = \phi_A \chi_A \sigma_A$

Solution:

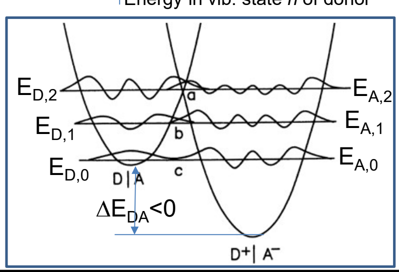
Energy in vib. state *m* of acceptor

$$k_{rt} = \frac{2\pi}{\hbar} \left| H_{21(el)} \right|^2 \sum_{n,m} \left[\frac{\exp\left(-E_{D,n}/kT\right)}{Z_1} \left| \langle \chi_{A,m} | \chi_{D,n} \rangle \right|^2 \times \delta\left(E_{A,m} - E_{D,n} + \Delta E_{DA}\right) \right] \times \langle \sigma_A | \sigma_D \rangle$$

Energy in vib. state *n* of donor

Recall Lecture 7:

$$k_{rt} = \frac{2\pi}{\hbar} \left| H_{21(el)} \right|^2 \sum_{n,m,u,w} \left[\begin{array}{l} \frac{\exp\left(-E_{n(1)}/kT\right)}{Z_1} \left| \langle \chi_{1m} | \chi_{1n} \rangle \right|^2 \times \\ \times \frac{\exp\left(-E_{u(2)}/kT\right)}{Z_2} \left| \langle \chi_{2w} | \chi_{2u} \rangle \right|^2 \times \\ \times \delta\left(\Delta E_{1nm} - \Delta E_{2uw}\right) \end{array} \right]$$



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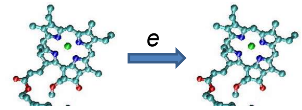
Electron transfer

Quantum-mechanical approach

Solution:

$$\boxed{\begin{aligned} k_{rt} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \times \\ \times \sum_{n,m}\left[\begin{aligned}&\frac{\exp\left(-E_{D,n}/kT\right)}{Z_1}\left|\left\langle \chi_{A,m} \mid \chi_{D,n}\right\rangle\right|^2 \\ &\times\delta\left(E_{A,m}-E_{D,n}+\Delta E_{DA}\right)\end{aligned}\right]\times \\ \times\left\langle \sigma_A \mid \sigma_D\right\rangle \end{aligned}}$$

Donor Acceptor



$\psi_D = \phi_D \chi_D \sigma_D$ $\psi_A = \phi_A \chi_A \sigma_A$

1. Interaction

   Recall: Born-Oppenheimer approximation, electron wavefunction separates from nuclear:

$$\left\langle \phi_A\chi_A\sigma_A \mid \tilde{H}_{AD} \mid \phi_D\chi_D\sigma_D\right\rangle = \left\langle \phi_A(\vec{r})\chi_A(\vec{R})\sigma_A(s) \mid \tilde{H}_{AD} \mid \phi_D(\vec{r})\chi_D(\vec{R})\sigma_D(s)\right\rangle$$
$$= \left\langle \phi_A \mid \tilde{H}_{AD} \mid \phi_D\right\rangle\left\langle \chi_A \mid \chi_D\right\rangle\left\langle \sigma_A \mid \sigma_D\right\rangle$$

$$\boxed{H_{AD(el)} \equiv \left\langle \phi_A \mid \tilde{H}_{AD} \mid \phi_D\right\rangle}$$

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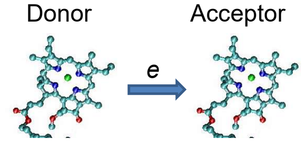
Electron transfer

Quantum-mechanical approach

1. Electron interaction:

$$\boxed{H_{AD(el)} \equiv \left\langle \phi_A \mid \tilde{H}_{AD} \mid \phi_D\right\rangle}$$

Donor Acceptor



$\psi_D = \phi_D \chi_D \sigma_D$ $\psi_A = \phi_A \chi_A \sigma_A$

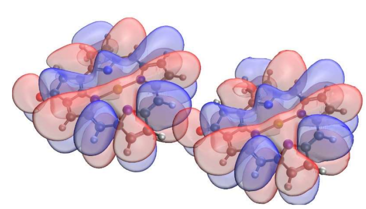
In the simplest form this is just electron tunneling:

$$H_{AD} \propto \sqrt{T} \propto \exp\left[-\left(\frac{\sqrt{2m_e\Delta\bar{E}_b}}{\hbar}\right)R_{DA}\right]$$

Since wavefunctions decay far from molecule exponentially, this term gives exponential dependence on the distance.

Example:

(Chl + $h\nu$) + Chl → Chl* + Chl → Chl$^+$ + Chl$^-$



Chl* (Chl LUMO) Chl$^-$ (Chl$^-$ LUMO)

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Electron transfer

Quantum-mechanical approach

1. Electron interaction:

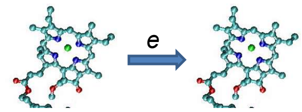
$$\boxed{H_{AD(el)} \equiv \langle \phi_A | \tilde{H} | \phi_D \rangle}$$

Alternative simplified approach:
what is the probability of finding an electron with wave function ϕ_D on the acceptor?

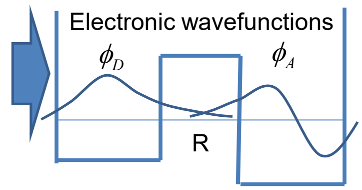
The probability would be:

$$H_{DA(el)} \propto \langle \phi_A | \phi_D \rangle$$

(Fourier trick)
Overlap of electronic wavefunctions on donor and acceptor.



$$\psi_D = \phi_D \chi_D \sigma_D \qquad \psi_A = \phi_A \chi_A \sigma_A$$



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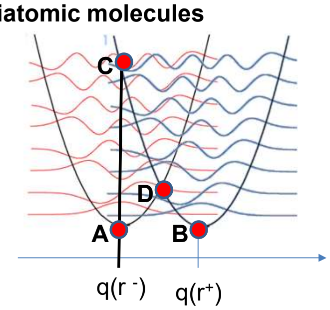
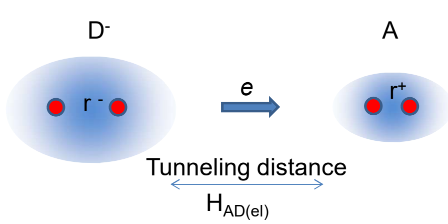
Electron transfer

The meaning of the nuclear term: homodimer of diatomic molecules





$$k_{rt} = \frac{2\pi}{\hbar} \left| H_{AD(el)} \right|^2 \times \\ \times \sum_{n,m} \left[\frac{\exp\left(-E_{D,n}/kT\right)}{Z_1} \left| \langle \chi_{A,m} | \chi_{D,n} \rangle \right|^2 \\ \times \delta\left(E_{A,m} - E_{D,n} + \Delta E_{DA}\right) \right] \times \\ \times \langle \sigma_A | \sigma_D \rangle$$

Left parabola: Nuclear energy dependence on distance between atoms with electron on donor
Right parabola: Nuclear energy dependence on distance between two atoms with missing electron (acceptor)

Points:
A: equilibrium distance with electron
B: equilibrium distance without electron
C: energy with electron on acceptor at equilibrium distance of molecule with electron; $E_C - E_A = \lambda$ (reorg. energy)
D: $r^- = r^+$ - electron on acceptor or donor will have the same energy

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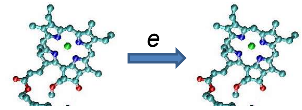
Electron transfer

Quantum-mechanical approach

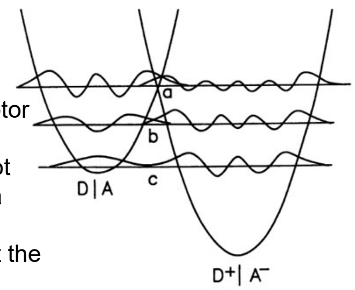
$$k_{rt} = \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \times \\ \times \sum_{n,m}\left[\frac{\exp\left(-E_{D,n}/kT\right)}{Z_1}\left|\langle \chi_{A,m} \mid \chi_{D,n}\rangle\right|^2 \times \delta\left(E_{A,m}-E_{D,n}+\Delta E_{DA}\right)\right] \times \\ \times \langle \sigma_A \mid \sigma_D \rangle$$

Donor Acceptor



$\psi_D = \phi_D \chi_D \sigma_D$ $\psi_A = \phi_A \chi_A \sigma_A$

2. Nuclear term:
   involves the overlap of nuclear wave functions
- Vibrational wave functions of the donor and acceptor must overlap along the reaction coordinate.
- these are vibrations of the Donor and Acceptor, not overlap between the ground and excited state of a single molecule (not like in energy transfer).
- Unlike classical physics, the transfer is possible at the lowest temperatures due to *tunneling*.
- FC's could be determined experimentally from the absorption spectra of donor and acceptor



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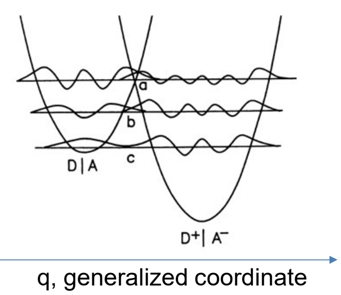
Electron transfer

Quantum-mechanical approach

Overlap of nuclear

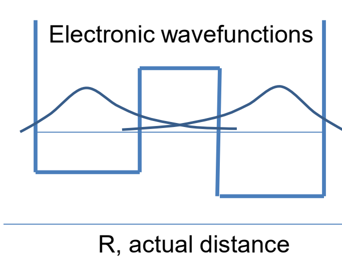
$\boxed{\langle \chi_{A,m} \mid \chi_{D,n} \rangle}$



and

electronic wavefunctions

$H_{DA(el)} \propto \langle \phi_A \mid \phi_D \rangle$



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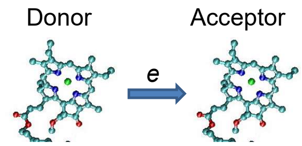
Electron transfer

Quantum-mechanical approach

$$
\begin{aligned}
k_{rt} &= \frac{2\pi}{\hbar}\left|H_{AD(el)}\right|^2 \times \\
&\times \sum_{n,m}\left[\begin{array}{l}\dfrac{\exp\left(-E_{D,n}/kT\right)}{Z_1}\left|\left\langle \chi_{A,m} \mid \chi_{D,n}\right\rangle\right|^2 \\ \times\delta\left(E_{A,m}-E_{D,n}+\Delta E_{DA}\right)\end{array}\right]\times \\
&\times\left\langle \sigma_A \mid \sigma_D \right\rangle
\end{aligned}
$$



$\psi_D = \phi_D \chi_D \sigma_D$ $\psi_A = \phi_A \chi_A \sigma_A$

3. Spin term
- Spin must be conserved!

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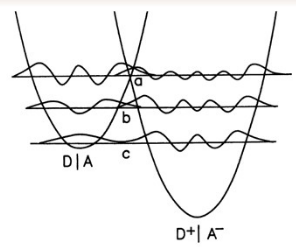
Electron transfer

Quantum-mechanical approach:

Nonadiabatic (or diabatic):
There are no mixed states, electron is on either donor or acceptor.



Electron transfer can occur:

a) Tunneling at the transition state:
The reactant (DA) and product (D^+A^-) have the same nuclear configuration, even when H_{AD} electron transfer can occur with high probability. However, transfer is temperature-dependent (even though electron tunneling is not)

b) Activated nuclear tunneling: nuclear surfaces are close enough and nuclear wavefunctions overlap. An electron can tunnel at energies below the transition point

c) Temperature-independent nuclear tunneling: Even at the lowest temperature there is overlap of nuclear wave functions.
For example: electron transfer in PS I RC at liquid helium temps has been shown experimentally to proceed at rates comparable to room temperature

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